Cyclotomic factorisation of P-192, P-224, P-256: what reaches the points, and point heights

Research Note: Cyclotomic Factorisation of P-192, P-224, P-256 — What Reaches the Points, and Point Heights

Date. 2026-10-11. **Status.** Derived and natively verified; no attack claimed, no ECDLP cost measured. Closes open items 4 and 5 of RESEARCH_NIST_SOLINAS_STRUCTURE.md and corrects one line there and in docs/ECDLP_CALIBRATION_PRIMES.md. **Code.** examples/nist_cyclotomic_point_heights.rs (Rust; num-bigint, curve_traits::arith, identity_certificate). **Evidence.** research/nist_cyclotomic_point_heights_20261011/: RESULTS.md (every table below, regenerated by the example), results.json, four identity_*.json certificates, cheon_cost.svg, structure.svg, REPORT.pdf, build.sh.

Curves are the FIPS 186 / SEC 2 curves P-192, P-224, P-256 ($y^2 = x^3 - 3x + b$, prime order n, generator G); the registry slugs for P-192 and P-256 are in docs/curves/registry.json.

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0. The question, and the one-paragraph answer

Does the cyclotomic structure of the NIST primes change the complexity of anything on the point group $E(F_p)$, and do lifted points have unusually small heights?

No, on both counts, and the note now says precisely why with numbers. Each prime is $f(2^{32})$ for a sparse Solinas polynomial f, and $f \pm 1$ factors in $\mathbb{Z}[t]$ with cyclotomic factors Φ_1, Φ_2, Φ_4 that force small prime factors of $p \pm 1$ (§1). **That includes P-256**, whose $p + 1 = 2^{96}(2^{64} + 1)(2^{96} - 2^{64} + 1)$ has largest prime factor 94 bits — contrary to the "no clean factorisation" line in two earlier documents. The structure lives in F_p^* and $F_{p^2}^*$. It reaches $E(F_p)$ only through the embedding degree, which is $(n-1)/10$, $(n-1)/3$, $(n-1)/3$ for the three curves (§2), so it never arrives. The integers that *do* act on the points — $n \mp 1$ (Cheon / Brown–Gallant), the twist order $N(\Phi_2(\pi))$, and the cyclotomic norms $N(\Phi_d(\pi))$ that factor $\#E(F_{p^k})$ — are not algebraically forced by the Solinas form; they are arithmetic accidents of each curve, and the one notable accident is P-256's $n - 1$, whose twelve prime factors are all at most 92 bits (§2.2). On heights (§3): the canonical lift $y^2 = x^3 - 3x + b$ has no integer or small-denominator rational point in a census of $9.8 \cdot 10^6$ candidates per curve (expected 10^{-22}–10^{-32}), the generator lifted xedni-style has canonical height $\hat{h} = (0.986\text{–}0.998) \cdot \log p$ with the Lang ratio $\hat{h} / \log|\Delta| \approx 1/6$, and a two-point lift has regulator $\approx \hat{h}(\tilde{G})\hat{h}(\tilde{R})$ with pairing cosine < 0.002: lifted points are as large and as independent as a random point can be. A point of $E(F_p)$ has a lift of naive height $\le H$ nats only when $|x| \le e^H$ or $|x - p| \le e^H$, i.e. the small-height points *are* the small-x factor base of RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md, whose reach is already bounded there.

Everything in §1–§3 is recomputed or verified by the example (polynomial identities by exact expansion and sealed as identity.certificate/v1 records; integer factorisations by product and Miller–Rabin on each factor; heights by the $4^{-k}\, h_x([2^k]P)$ limit, which agrees with PARI's ellheight to the printed digits). The one fact below that is a PARI cross-check only, not reproduced natively, is labelled as such (§3.1).

1. Algebraic (cyclotomic) factorisation of $p \pm 1$

With `t = 2^32`:

curve	f(t)	f − 1	f + 1
P-192	$t^6 - t^2 - 1$	$t^6 - t^2 - 2$, irreducible	$t^2 \cdot \Phi_1 \Phi_2 \Phi_4 = t^2(t^4 - 1)$
P-224	$t^7 - t^3 + 1$	$t^3 \cdot \Phi_1 \Phi_2 \Phi_4 = t^3(t^4 - 1)$	$t^7 - t^3 + 2$, irreducible
P-256	$t^8 - t^7 + t^6 + t^3 - 1$	$\Phi_1 \Phi_2 \cdot (t^6 - t^5 + 2t^4 - t^3 + 2t^2 + 2)$	$t^3 \cdot \Phi_4 \cdot (t^3 - t^2 + 1)$

Certificates (`identity.certificate/v1`, 64 points over `F_{2^61−1}`, degree bound 8, so a false identity passes with probability `< 2^{-61·64+…}`; the exact `Z[t]` expansion in the example is the independent check): P-192 `f + 1` `IDC1h52442d0716847371`; P-224 `f − 1` `IDC1h7d04feed32d16d24`; P-256 `f − 1` `IDC1h8c257e9ef061103a`; P-256 `f + 1` `IDC1h923abc2d4d3a80f4`.

The integer consequence (full factorisations, discovered with PARI 2.17 and verified natively; largest prime factor in bits):

curve	p − 1	largest	p + 1	largest
P-192	2 · 59 · 149309 · 11393611 · 108341181769254293 · 288626509448065367648032903	88	2^64 · 3 · 5 · 17 · 257 · 641 · 65537 · 274177 · 6700417 · 67280421310721	46
P-224	2^96 · 3 · 5 · 17 · 257 · 641 · 65537 · 274177 · 6700417 · 67280421310721	46	2 · 41 · 2671 · 10637177 · 78987407 · 1465031779237846920418054071651647727462566025 29	157
P-256	2 · 3 · 5² · 17 · 257 · 641 · 1531 · 65537 · 490463 · 6700417 · 835945042244614951780389953367877943453916927241	160	2^96 · 7 · 274177 · 67280421310721 · 113183089279739419314049141 03	94

Reading the algebra off the numbers: `Φ_1Φ_2Φ_4(2^32) = 2^128 − 1 = 3·5·17·257·641·65537·274177·6700417·67280421310721` is the whole odd part of P-224's `p − 1` and of P-192's `p + 1`; for P-256, `Φ_1Φ_2(2^32) = 2^64 − 1` contributes `3·5·17·257·641·65537·6700417` to `p − 1` and `Φ_4(2^32) = 2^64 + 1 = 274177 · 67280421310721` together with `2^96 − 2^64 + 1 = 7 · 113183089279739419314049141 03` make up `p + 1`. The irreducible cofactors ($t^6 - t^2 - 2$, $t^7 - t^3 + 2$, the P-256 sextic) carry the large primes; their factorisations are accidents.

Correction. `docs/ECDLP_CALIBRATION_PRIMES.md` §2 and `RESEARCH_NIST_SOLINAS_STRUCTURE.md` §2 listed the 5-term P-256 shape as "no clean cyclotomic factorisation". The 5-term shape $2^n - 2^a + 2^b + 2^c - 1$ has `p + 1 = 2^c(2^{n−c} − 2^{a−c} + 2^{b−c} + 1)`, and for P-256's exponents `(a, b, c) = (224, 192, 96)` the bracket is `(2^64 + 1)(2^96 − 2^64 + 1)`. This is specific to P-256's exponents (it needs `n − c = a − c + b − c`, i.e. `n + c = a + b`, which `256 + 96 = 224 + 192` satisfies), so the *family* row stays "no forced factor" but the NIST anchor does not. Both documents are amended in this change.

Where it acts. Pohlig–Hellman in `F_p^*` costs about the square root of the largest factor of `p − 1`: `2^44` for P-192, `2^23` for P-224, `2^80` for P-256 (where NFS at 256 bits is cheaper anyway). In `F_{p²}^*`, order `(p − 1)(p + 1)`, the largest factors are 88, 157 and 160 bits. None of this is an ECDLP statement; it says these primes must not double as finite-field DLP moduli, which `RESEARCH_NIST_SOLINAS_STRUCTURE.md` §5 already concluded for P-224.

2. What reaches the point group

2.1 Embedding degree — the only bridge, and it is closed

`ord_n(p)`, computed exactly from the complete factorisation of `n − 1`:

curve	ord_n(p)	bits
P-192	(n − 1)/10	189
P-224	(n − 1)/3	223
P-256	(n − 1)/3	255

This replaces the lower bounds in the repository (`> 10⁸` in `docs/guides/P256_PROPERTIES.md`, `> 1000` in the P-192 walk) with exact values. The smoothness of `p ± 1` sits in `F_p^*` and `F_{p²}^*`; the pairing lands in `F_{p^k}` with `k ≈ 2^{188}–2^{254}`. There is no path.

2.2 `n ∓ 1` — Cheon / Brown–Gallant, the one place the points feel a factorisation

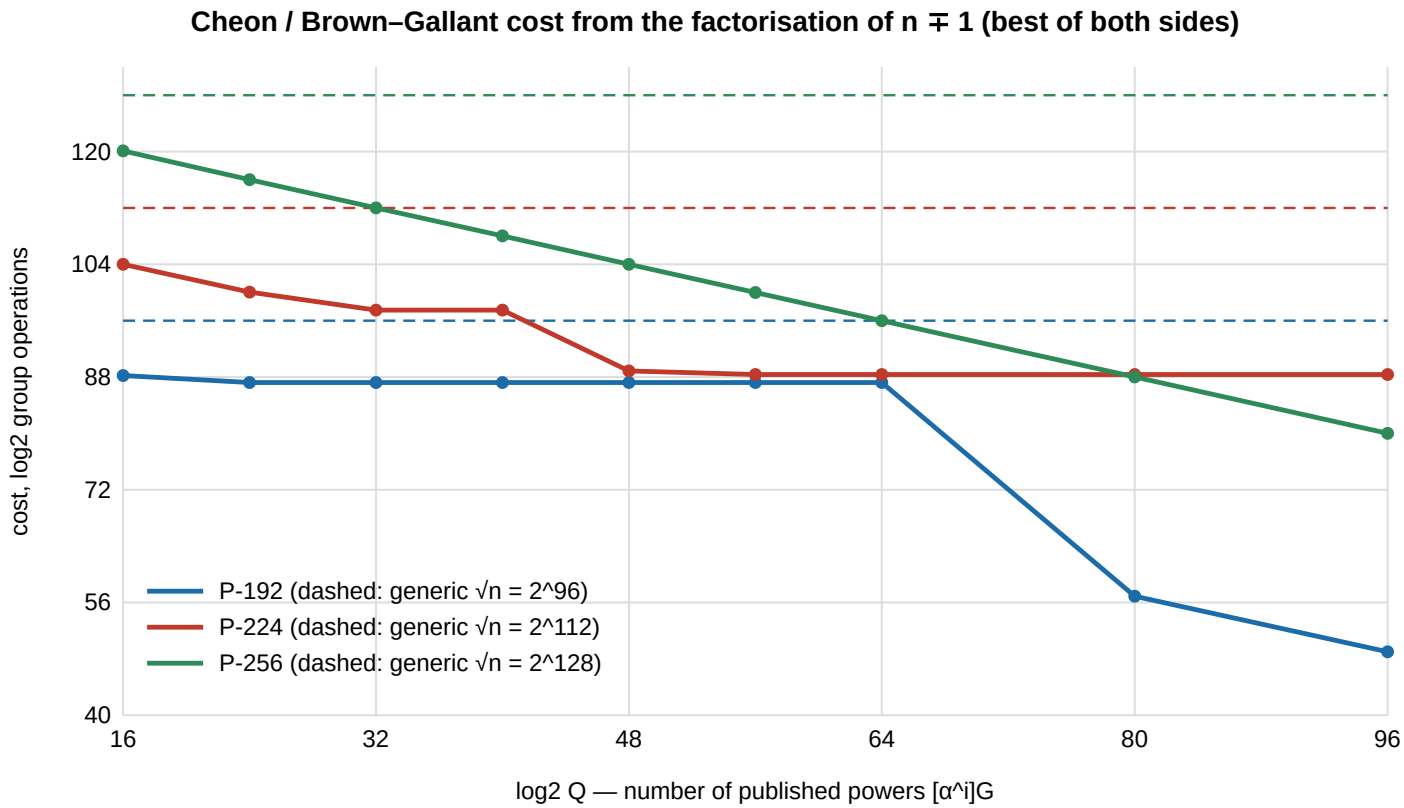
`n = p + 1 − t` with a 95/112/127-bit trace, so the Solinas form forces nothing about `n ∓ 1`. Complete factorisations:

curve	n − 1	largest	n + 1	largest
P-192	2^4 · 5 · 2389 · 95646823139138600591956 69 · 34338591793161886821199869 11	92	2 · 3 · 41 · 116696426742853568300653 · 21865856936057981464360692617 8439	108
P-224	2² · 3⁶ · 5 · 17 · 2153 · 505206062588758187074708601532876667009176960999333893515 07	196	2 · 7 · 19 · 641707280681 · 157943015469068292359817151928968170360745244072702747	177
P-256	2^4 · 3 · 71 · 131 · 373 · 3407 · 17449 · 38189 · 187019741 · 622491383 · 1002328039319 · 26247475503338692784167739 53	92	2 · 5 · 1879 · 176337611 · 34946779280882916835155272231706129710560967816144871596921775673	215

P-256's `n − 1` is the outlier: twelve prime factors, none above 92 bits, smooth part (everything but the largest prime) `2^{164}`. For a random 256-bit integer the probability that the largest prime factor is below `2^{92}` is Dickman `ρ(256/92) ≈ 0.03`, so this is a 1-in-30 accident, not a design artefact. P-224's smooth part is `2^{28}`, P-192's `2^{100}` but with nothing usable between `2^{17.5}` and `2^{83}`.

Where it matters: a protocol that publishes `[α^i]G` for `i ≤ Q` (q-SDH setups, powers-of-tau; **not** ECDSA or ECDH) lets Cheon solve for `α` in `√(n/d) + √d` operations for any `d | n − 1` with `d ≤ Q` (or `d | n + 1` with `2d ≤ Q`). Best divisor and cost in bits (`√n` generic is 96 / 112 / 128):

log2 Q	P-192	P-224	P-256
16	88.2	104.0	120.1
32	87.2	97.5	112.0
48	87.2	88.9 (n + 1 side)	104.0
64	87.2	88.4 (n + 1 side)	96.0
80	56.9 (n + 1, d = 2^{78})	88.4	88.0
96	49.0	88.4	80.0



Cheon / Brown–Gallant cost against published powers

P-256 gives up exactly `log2(Q)/2` bits for every `Q` up to `2^{96}`, because `n − 1` has a divisor within a factor 1.05 of every power of two in that range; P-224 is capped at 23.6 bits of saving however many powers are published. This is consistent with, and extends, the `Q ≤ 2^{32}` / `Q ≤ 2^{64}` columns of RESEARCH_TORSION_AUXILIARY_INPUTS.md §“What a leak would buy”, which stopped at trial division plus bounded rho. It is also the correction to `docs/guides/ELLIPTIC_CURVE_ATTACKS_AND_TYPES_GUIDE.md` §3.8, which wrote the Cheon parameter as a divisor of `p − 1`; it divides the *group order* side `n ∓ 1`, which is why the `p ± 1` smoothness of §1 buys Cheon nothing.

2.3 Cyclotomic factorisation *of the points*: `#E(F_{p^k}) = ∏_{d | k} N(Φ_d(π))`

The honest meaning of “cyclotomic factorisation over the points” is the Frobenius one. With `π² − tπ + p = 0`, `#E(F_{p^k}) = ∏_{d|k} N_d`, `N_d = N_{Q(π)/Q}(Φ_d(π))`, verified against `p^k + 1 − V_k(t, p)` for all `k ≤ 12`; `N_1 = n`, `N_2 = p + 1 + t` is the twist order. Trial division plus bounded rho on each `N_d` (`P` probable prime, `C` unsplit composite, subscripts are bit lengths):

d	P-192	P-224	P-256
2 (twist)	23 · P95 · P95	3^2 · 11 · 47 · 3015283 · 40375823 · 267983539294927 · P118	3 · 5 · 13 · 179 · P241
3	2^2 · 10453 · 551527921513 · C330	2^2 · 719746610461 · C407	2^2 · 606116377 · 635514031 · C452
4	5 · 13 · P378	C448	13 · 10657 · 318281 · 842272153 · C447
6	2^2 · 181 · 223 · C367	2^4 · 3^2 · 5^2 · 17^2 · 877 · C419	2^2 · 3^3 · 43 · 76213 · 3655760077 · P452
12	2^2 · 13 · C763	2^2 · 16674781 · C871	2^2 · 373 · 1908157 · C993

(The twist rows are the complete factorisations from §2.2's run; the others are what budget `2^{16}` rho splits.) Nothing here is small enough to matter: every N_d with $d \geq 3$ keeps a cofactor of at least 330 bits, and the only $d \geq 3$ subgroup an attacker could even address is the one $E(\mathbb{F}_p)$ itself sits in. The twist orders restate the known twist-security picture — P-224's twist has a 118-bit largest factor (an invalid-curve attack recovers the key in about 2^{59} work if points are not validated), P-192's 95 bits, P-256's 241 bits.

2.4 Trace and CM discriminant, for completeness

t: 89 · 179 · 383 · 6196490257 · 835990267859 (P-192), 17626237 · 55872356921 · 4806058943693873 (P-224), 113 · 2521 · 12713 · 3450079 · 7138040046852502726273 (P-256). $4p - t^2$: 5 · 11 · 31 · P184, 3^3 · 29 · 79 · 7523 · 40927 · 11549194661 · P149, 3 · 5 · 456597257999 · 1428624589419343516204097 · P136. The fundamental discriminants are huge and the class numbers $\approx 2^{125}$-scale, as `docs/guides/P256_PROPERTIES.md` §2 and the P-192 walk already record; no new isogeny-graph fact, and no canonical graph changes.

3. Point heights

3.1 The canonical lift has no small points

$E_b/\mathbb{Q} : y^2 = x^3 - 3x + b$ with the integer $b \in [0, p)$; $\Delta = -432\,(b-2)(b+2)$, $\log_2|\Delta| = 390.0 / 455.7 / 517.8$. Census of integer x with $|x| \leq 10^6$ and rational $x = u/d^2$ with $2 \leq d \leq 40$, $|u| \leq 10^5$ (residue filter mod 64, 63, 65, 11, then exact square test):

curve	candidates	after filter	points	expected
P-192	9 800 040	21 601	0	$2.0 \cdot 10^{-22}$
P-224	9 800 040	29 905	0	$2.3 \cdot 10^{-27}$
P-256	9 800 040	70 300	0	$4.8 \cdot 10^{-32}$

Zero, as expected: each candidate is a square with probability about $b^{-1/2}$. The torsion of $E_b(\mathbb{Q})$ is trivial (PARI `elltors`, not reproduced natively). **PARI cross-check only:** $(b \mp 2)$ factor as products of 2–9 primes with no repeated factor for all three curves, so E_b is semistable away from 2 and 3, the conductor is $\approx |\Delta|$ and the Szpiro ratio $\log|\Delta_{\min}| / \log N$ is 1.000 / 1.000 / 1.002. Under the Lang–Silverman conjecture a non-torsion point then has $\hat{h} \geq c \cdot \log|\Delta_{\min}|$ with c absolute;

the smallest-height point on E_b is therefore of height at least some constant times 270–360 nats, and no point of E(F_p) has a lift anywhere near the O(1) heights that would make a sieve possible.

3.2 Lifted generator: ĥ = log p, Lang ratio 1/6

xedni-style one-point lift: E_1 : y² = x³ − 3x + B_1 through G̃ = (G_x, G_y) ∈ Z², B_1 = G_y² − G_x³ + 3G_x (566 / 671 / 765 bits). ĥ := lim 4^{-k} h_x([2^k]P) (the x-coordinate normalisation, twice Silverman's AEC ĥ), computed in Jacobian coordinates over Z; the sequence is stationary to four decimals from k = 2:

curve	ĥ(G̃) nats	bits	ĥ / log p	log|Δ(E_1)|	ĥ / log|Δ|	h_x(G̃) − ĥ(G̃)
P-192	131.289	189.4	0.9865	790.5	0.1661	−0.549
P-224	153.066	220.8	0.9858	935.6	0.1636	+1.864
P-256	177.124	255.5	0.9982	1065.5	0.1662	−0.549

PARI ellheight on the same curves gives 131.289, 153.066, 177.124: the native limit and the archimedean-series computation agree. Two readings. (i) ĥ ≈ h_x to within two nats although Silverman's bounds allow a discrepancy of order log|Δ|/12 ≈ 70–90 nats: the lift has no cancellation to offer. (ii) ĥ / log|Δ_min| ≈ 1/6 in this normalisation (1/12 in Silverman's), against a Hindry–Silverman lower bound of order 10^{-15} for Szpiro ratio 1 — the lifted generator is nowhere near a small point of its curve.

3.3 Two-point lift: independent, orthogonal

E_2 : y² = x³ + Ax + B, A, B ∈ Q, through G̃ and R̃, R = [k]G, k = 0x1234567890abcdef:

curve	ĥ(G̃)	ĥ(R̃)	⟨G̃, R̃⟩	cos ∠	regulator
P-192	196.91	196.28	0.140	0.0007	3.87·10⁴
P-224	229.73	229.77	0.297	0.0013	5.28·10⁴
P-256	263.97	264.03	0.457	0.0017	6.97·10⁴

(Heights on E_2 exceed those on E_1 by the factor ≈ 1.48 because clearing the 189–253-bit denominator of A rescales the model; the canonical height is a model invariant only up to that rescaling.) Xedni needs the lifted points to be *dependent*; the regulator is as large as the product of the heights allows and the pairing is zero to three decimals. This is Jacobson–Koblitz–Silverman–Stein–Teske (2000) seen in the numbers, and it is the same obstruction RESEARCH_IC_LEADS_AND_CONNECTIONS_20261010.md §(d) states through the parallelogram law ĥ(P + Q) + ĥ(P − Q) = 2ĥ(P) + 2ĥ(Q): the sum of two height-log p points has height ≈ 2 log p, so no height-bounded factor base is closed under the group law.

3.4 "Smaller point heights" quantified

A point P ∈ E(F_p) has a lift to some Weierstrass model over Q of naive height ≤ H nats only if its x-coordinate has a representative of absolute value ≤ e^H, i.e. x ∈ [0, e^H] ∪ [p − e^H, p). The fraction of the group with that property is 2e^H / p:

H (bits)	fraction of $E(\mathbb{F}_p)$, P-256	what it is
64	2^{-191}	$\delta = 1/4$ small-x base ($p^{1/4}$): the threshold at which §2.1 of the breakthrough note needs $O(1)$ lattice reductions per trial
85	2^{-170}	$p^{1/3}$: the Bernstein–Lange advice size
128	2^{-127}	$\sqrt{p}$: a base as large as rho's running time
255	≈ 1	every point: height $\log p$, the measured $\hat{h}(\tilde{G})$

So the only points with "smaller heights" are the small-x points, their usefulness is exactly the Coppersmith-reach question already closed in `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §2, and the cyclotomic shape of p does not enter: the height of a lift depends on the *size* of x, not on the arithmetic of p.

4. Decision and what is not here

- **No change to any ECDLP cost, scoreboard row or attack-matrix entry.** `docs/ECDLP_ATTACK_MATRIX.md`, `docs/index-calculus-scoreboard.html` and the curve diagrams were checked; nothing in this note is graphable there.
- **Two factual corrections** land with this note (P-256's $p + 1$; the Cheon parameter divides $n \mp 1$, not $p - 1$), and `docs/guides/P256_PROPERTIES.md`'s embedding-degree lower bound is superseded by the exact value above (left in place, as it is a frozen probe report; this note is the current statement).
- **Not done:** the $4p - t^2$ cofactors `P184`, `P149`, `P136` and the `N_d` cofactors for $d \geq 3$ are probable primes / unsplit composites; nothing here depends on them being prime. The conductor and torsion facts in §3.1 are PARI-only. A native ECM would let the example split the `C140`–`C216` cofactors its rho leaves (the frozen lists cover them).

5. Reproduce

```
cargo run --release --example nist_cyclotomic_point_heights -- \
    --out research/nist_cyclotomic_point_heights_20261011        # ≈ 70 s
sh research/nist_cyclotomic_point_heights_20261011/build.sh       # REPORT.pdf
```

Cross-references. `RESEARCH_NIST_SOLINAS_STRUCTURE.md` (mechanism, P-224 worked out, calibration rows), `RESEARCH_NIST_SOLINAS_EXPERIMENTS.md` (measured Pohlig–Hellman in $\mathbb{F}_p^*$, twist leakage), `RESEARCH_TORSION_AUXILIARY_INPUTS.md` (Cheon on the repository's rho reference), `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` (small-x bases and Coppersmith reach), `RESEARCH_IC_LEADS_AND_CONNECTIONS_20261010.md` (the height obstruction as a licensing datum), `docs/guides/P256_PROPERTIES.md` (twist, CM discriminant, earlier embedding-degree probe).

Figures

Where the cyclotomic structure of the NIST primes acts — and where it does not

P-192: $p = t^6 - t^2 - 1$ **P-224: $p = t^7 - t^3 + 1$** **P-256: $p = t^8 - t^7 + t^6 + t^3 - 1$**

$t = 2^{32}$; $\Phi_1\Phi_2\Phi_4(t) = 2^{128} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 274177 \cdot 6700417 \cdot 67280421310721$ (largest prime 46 bits)

Multiplicative groups F_p^*, $F_{p^2}^*$ — algebraic, forced by $f \pm 1$ in $Z[t]$

$p + 1 = t^2 \cdot \Phi_1\Phi_2\Phi_4 = 2^{64}(2^{128} - 1)$
largest prime 46 bits — forced
$p - 1$: $t^6 - t^2 - 2$ irreducible, largest 88 bits

$p - 1 = t^3 \cdot \Phi_1\Phi_2\Phi_4 = 2^{96}(2^{128} - 1)$
largest prime 46 bits — forced
$p + 1$: $t^7 - t^3 + 2$ irreducible, largest 157 bits

$p + 1 = t^3 \cdot \Phi_4 \cdot (t^3 - t^2 + 1)$, largest 94 bits — forced
$p - 1 = \Phi_1\Phi_2 \cdot$(sextic): $2^{64} - 1$ forced, largest 160 bits
corrects “no clean factorisation” in two earlier docs

Pohlig–Hellman in F_p^*: $\approx 2^{44}$ (P-192, via $p - 1$), 2^{23} (P-224), 2^{80} (P-256). Finite-field DLP only.
In $F_{p^2}^*$ the order is $(p - 1)(p + 1)$: largest factors 88 / 157 / 160 bits.
Deployment rule it proves: never reuse these primes as DSA / Schnorr / pairing moduli.

only bridge: embedding degree ord_n(p)
$= (n - 1)/10$, $(n - 1)/3$, $(n - 1)/3 \rightarrow$ 189 / 223 / 255 bits: closed ✕

Point group $E(F_p)$ — what it actually feels (not forced by the Solinas form)

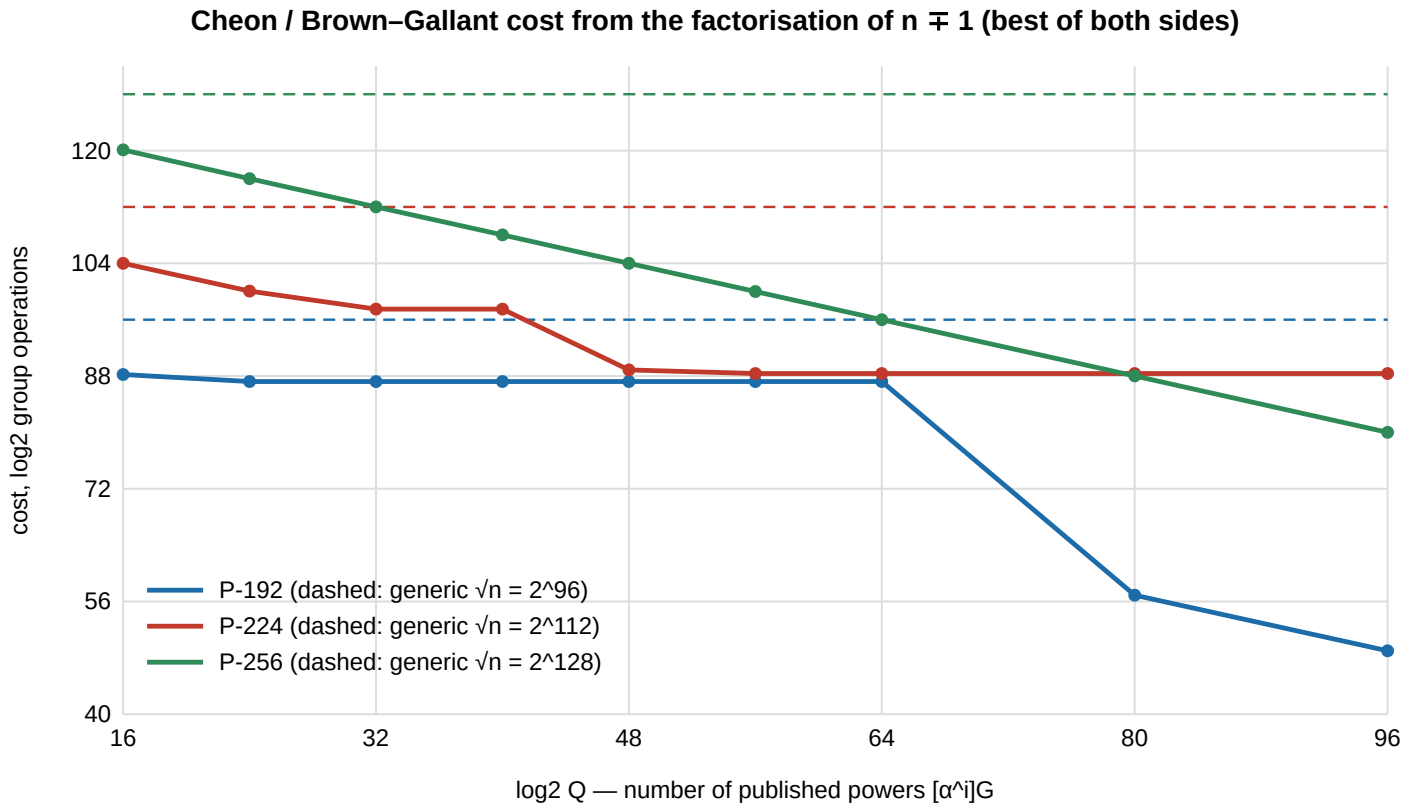
$n \mp 1$ (Cheon / Brown–Gallant, needs published $[\alpha^l]G$)
P-192: $n - 1 = 2^4 \cdot 5 \cdot 2389 \cdot P83 \cdot P92$ — smooth part $2^{17.5}$ usable
P-224: $n - 1 = 2^2 \cdot 3^6 \cdot 5 \cdot 17 \cdot 2153 \cdot P196$ — smooth part 2^{29}
P-256: $n - 1 = 2^4 \cdot 3 \cdot 71 \cdot 131 \cdot 373 \cdot 3407 \cdot 17449 \cdot 38189 \cdot 187019741 \cdot 622491383 \cdot 1002328039319 \cdot P92$
P-256: every power of two up to 2^{96} has a divisor of $n - 1$ within 5 % — cost $\sqrt{(n/Q)}$ (see cheon_cost.svg)

Frobenius cyclotomic factorisation of the points: $\#E(F_{p^k}) = \prod_{d|k} N(\Phi_d(\pi))$
N_2 = twist order: $23 \cdot P95 \cdot P95 / 3^2 \cdot 11 \cdot 47 \cdot 3015283 \cdot 40375823 \cdot 267983539294927 \cdot P118 / 3 \cdot 5 \cdot 13 \cdot 179 \cdot P241$
N_d, $d \geq 3$: every one keeps an unsplit cofactor of ≥ 330 bits — no small extension subgroup to address

Heights of lifts to Q (the “smaller point heights” question)
canonical lift $y^2 = x^3 - 3x + b$: 0 points in $9.8 \cdot 10^6$ small candidates per curve (expected $10^{-22} \ldots 10^{-32}$)
lifted generator: $\hat{h}(\tilde{G}) = 0.99 \cdot \log p$, $\hat{h}/\log|\Delta| \approx 1/6$; two-point lift: pairing cos < 0.002, regulator $\approx \hat{h}(\tilde{G}) \cdot \hat{h}(\tilde{R})$
a point has a lift of height $\leq H$ only if $|x| \leq e^H$: fraction $2e^H/p$ of the group = the small-x factor base, nothing else

research/nist_cyclotomic_point_heights_20261011 — derived + natively verified; PARI used for discovery only

Where the cyclotomic structure acts, and where it does not



Source: results.json here. Exact divisors of n − 1, n + 1; cost √(n/d) + √d, d ≤ Q (resp. 2d ≤ Q). Three curves, exact values, no uncertainty.

Cheon / Brown–Gallant cost against published powers